Four-person build plan

Two circuits, two pairs | Assign one owner to each task before starting

Build A on the small breadboard and B on the large breadboard. Persons 1 and 3 assemble hardware while Persons 2 and 4 prepare the matching code. Then each pair tests its own circuit. Use the MMA7361 version of the supplied code package.

Owner / name	Responsibility	Finished result
Person 1 _____	B - station hardware Schematic: page 2; task: page 4	Labelled station circuit; stable sensor mount; wiring checked
Person 2 _____	B - station software and tests Task: page 5	Wall-sit/plank selection, timers, OLED/Serial output and saved test log
Person 3 _____	A - wearable hardware Schematic: page 3; task: page 6	MMA7361 wired and mounted; GY-MAX30102 electrical checks recorded
Person 4 _____	A - wearable software and tests Task: page 7	Movement trial, treadmill timer, HR trial if checks pass, and saved test log

Parts allocation

Circuit A - small breadboard	Circuit B - large breadboard
Uno R4 Minima; MMA7361; GY-MAX30102; 1 bare button; USB cable	Uno R3; HC-SR04; OLED; 2 bare buttons; USB cable

Controller models follow your pitch. Confirm labels before uploading. No Grove shield is needed. Leave the B500K knob and four-digit display out of the first build. Select A exercises through Serial Monitor. Each circuit has its own power and ground; there are no inter-board wires.

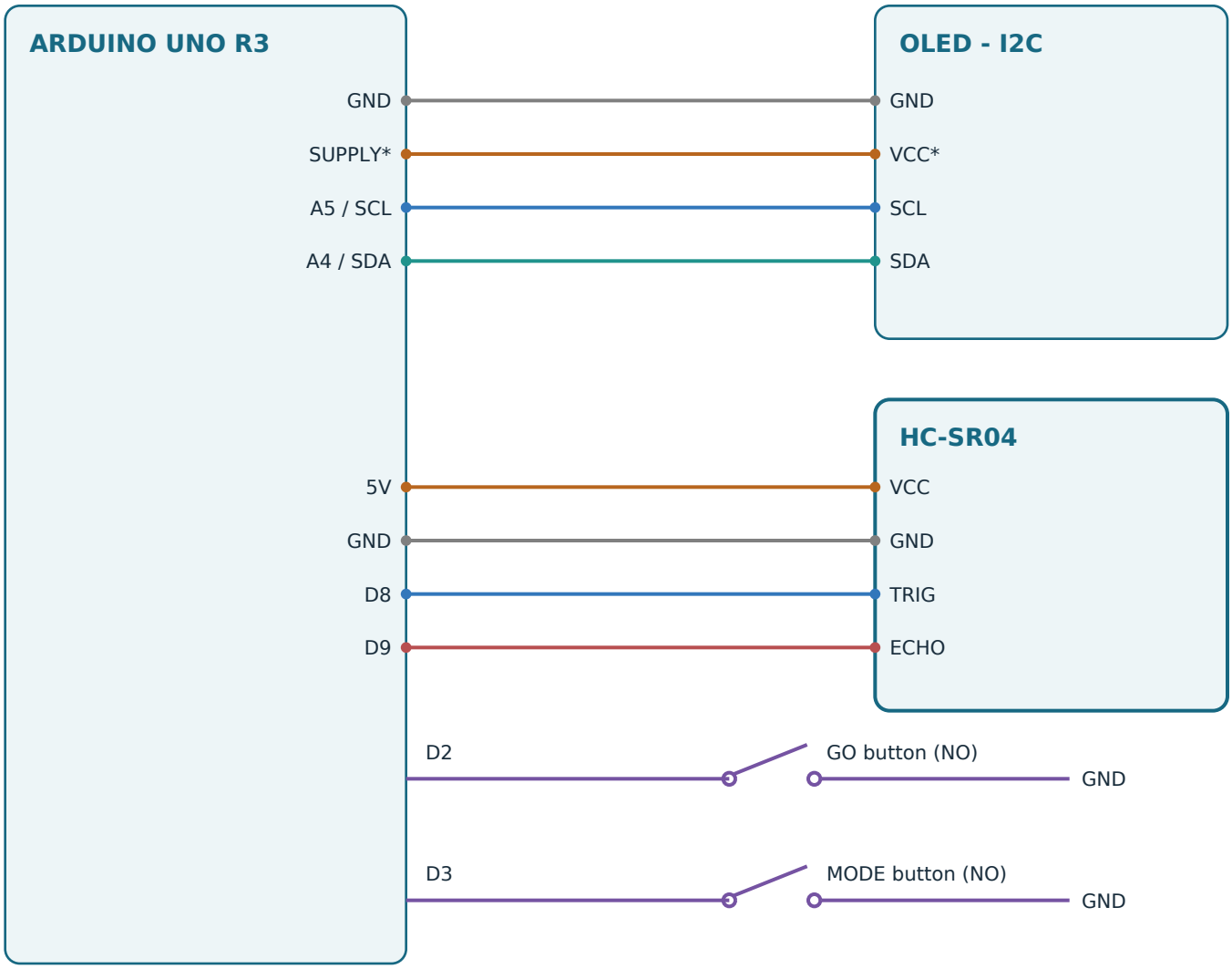
Exercise allocation

B: wall sit and plank.
A: treadmill (warm-up + 15-minute cardio timer and HR), thoracic rotations, airplane, chest press, glute bridges, lat pulldown, farmer's carry, hip flexor stretch and cat-cow. Automatic repetition counting is a trial feature for selected movements; the remaining activities use timers/manual confirmation.

Handoffs: Person 1 hands B to Person 2 after wiring checks; Person 3 hands A to Person 4 after sensor checks. Persons 2 and 4 save the data and choose repeatable demonstrations. Bring both circuits, labelled wiring and the test results to the progress presentation.

Schematic B - station

Large breadboard | Uno R3 | Wall sit and plank



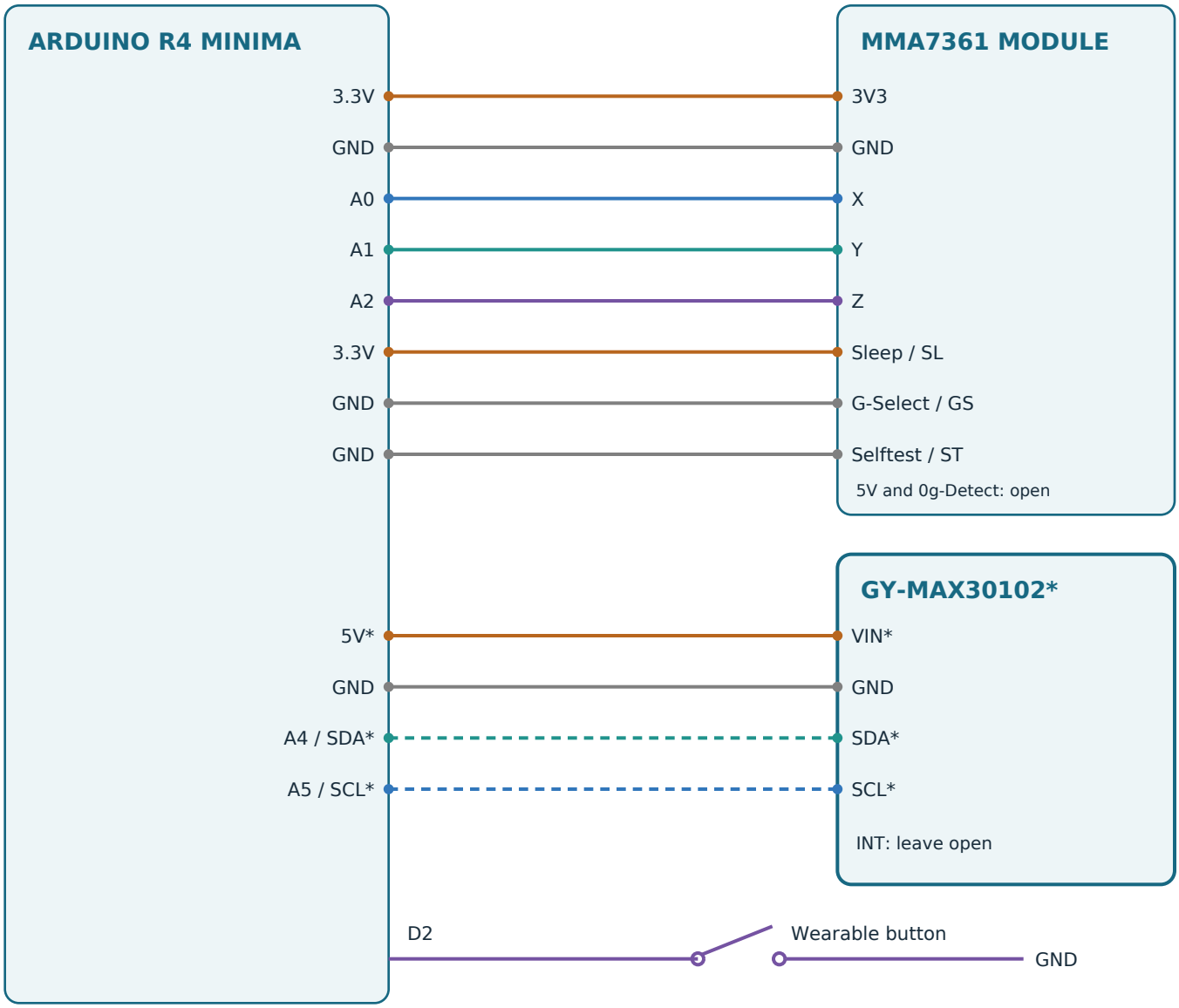
This is an electrical connection diagram, not a physical pin layout. Follow the printed labels. In your OLED photo the front header reads GND, VCC, SCL, SDA.

Connection	Build instruction
OLED supply*	Use its verified module supply/interface rating. If documented for 5V use Uno 5V. If unknown, leave OLED unplugged and start with ENABLE_OLED=0 using Serial Monitor.
GO and MODE	Each bare button connects its pin to GND only when pressed. INPUT_PULLUP is enabled in the code; external resistors are unnecessary.
Power	USB powers the Uno. All B components share Uno GND. Keep its 5V rail separate from A.

OLED: ENABLE_OLED=1; OLED_SH1106=1 (0 for SSD1306). OLED_ADDRESS=0x3C initially; printed 0x78/0x7A means 0x3C/0x3D. Controller and voltage rating remain unconfirmed.

Schematic A - wearable

Small breadboard | Uno R4 Minima | MMA7361 and GY-MAX30102



MMA7361: connect Arduino 3.3V to module 3V3 and Sleep. Ground GS and ST. Leave the module 5V and 0g-Detect pins open. Do not apply 5V to its control pins.

GY-MAX30102*: proposed wiring for the common five-pin regulated/level-shifted GY board. Your photo is consistent with that layout, but cannot prove its electrical connections. Person 3 must complete the power/interface check on page 6 before connecting the dashed SDA/SCL wires. If incompatible, add a suitable I2C level shifter or leave HR disconnected.

Power A through USB. All A modules share its ground. Do not connect A's 5V and 3.3V rails together. The optional potentiometer is omitted: USE_SELECTOR_POT=0. The code initially has ENABLE_MAX30102=0; Person 4 enables it after Person 3's check.

Person 1 - build station B

Name: _____ | Hardware owner | Work with Person 2

Your task

Assemble the station from page 2, secure the ultrasonic sensor, and give Person 2 a circuit with checked wiring. Person 2 owns the program and timing tests; you own connections and mounting.

Build steps

1. Put the Uno R3 beside the large breadboard. Keep USB unplugged while wiring.
2. Connect HC-SR04: VCC to 5V, GND to GND, TRIG to D8 and ECHO to D9.
3. Wire GO from D2 to GND and MODE from D3 to GND through normally open contacts.
4. Connect OLED SDA to A4 and SCL to A5. Connect its power only after confirming its module rating; otherwise leave it unplugged for the first Serial-only test.
5. Label the wires and fix the ultrasonic sensor to a stand so it cannot move during a trial.

Breadboard checks

Each numbered a-e row is one connected group; f-j is another. The centre channel separates them. Each inline module pin must occupy a different group. A four-pin header placed across one connected row would short its pins. Check split power rails and the button's internally connected legs.

Check	What you record	Pass / fail
Unpowered wiring review	Correct pins, no 5V-GND short, and button contacts close only when pressed	_____
Distance demonstration	With Person 2, move a flat card to 20cm and 40cm from the sensor face; record both readings	_____
Mount stability	Repeat a reference position without the sensor stand shifting	_____
Body target	Aim at torso/hip, not the wall behind the participant; confirm echoes remain usable	_____

For the card test, a proposed bench target is within +/-2cm at each distance. This is a team test target, not a result already achieved. For body trials, prioritize repeatable echoes and a stable mount.

Handoff to Person 2

Give them: one labelled circuit, a clear photo, the OLED power/driver information if known, and your measurements. Tell them any wiring or mounting issue that remains.

Done when: the wiring checks pass and Person 2 can repeatedly read the ultrasonic distance.
Owner sign-off: _____ Checked by: _____

Person 2 - program and test B

Name: _____ | Station software owner | Work with Person 1

Prepare while Person 1 builds

1. Open HoldStation/HoldStation.ino, with PrototypeCore.h in the same folder.
2. Select Arduino Uno and its COM port. Install Arduino AVR Boards and the U8g2 library.
3. Start with ENABLE_OLED=0 if the OLED rating is unconfirmed. Otherwise try OLED_SH1106=1; use 0 for SSD1306 if needed.
4. Upload. Open Serial Monitor at 115200 baud, Newline. Send help. Close Serial Monitor before using the capture script on the same port.

Calibrate and operate

Tap MODE to select wall sit or plank while IDLE. Hold the chosen reference posture, hold MODE for >=1.2s and release, then stay still for the 2-second calibration. Keep reference distance between 12 and 150cm. Tap GO; the program starts after a 3-second countdown. Tap to pause/resume; hold GO and release to finish.

Active is elapsed running time. In band is time with usable echoes within +/-4cm of the reference, after a short stable entry. It is a distance proxy, not a full-body posture assessment. Changing exercise modes clears the calibration.

Test	Expected behavior	Record
30-second run	Active time agrees with stopwatch within +/-1s; exclude countdown	_____
10-second pause	Active and in-band totals stay unchanged	_____
Move outside band	In-band time stops within about 0.5s; active time continues	_____
Remove target / no echo	No valid echo is shown; no stale in-band credit	_____
Change wall / plank	Program requires a new reference calibration	_____
Repeat real hold	Run each hold three times; save time, dropouts and setup notes	_____

Save your evidence

Use tools/capture_serial.py to save station.csv, or copy labelled Serial results. Record reference distance, tolerance and sensor position. Keep one repeatable wall-sit or plank demonstration for the presentation.

Done when: the timer, pause and dropout tests pass and a session SUMMARY is saved.
Owner sign-off: _____ Checked by: _____

Person 3 - build wearable A

Name: _____ | Hardware owner | Work with Person 4

Build the motion circuit first

1. Use page 3. Connect MMA7361 3V3 and Sleep to Arduino 3.3V; connect its GND, GS and ST to GND.
2. Connect X/Y/Z to A0/A1/A2. Leave the MMA7361 5V and 0g-Detect terminals open.
3. Connect the bare button between D2 and GND. Leave the B500K selector out of this build.
4. Ask Person 4 to run SerialDiagnostics. Keep the sensor still, then slowly tilt it; its analog outputs should change repeatably.
5. Prepare a secure removable sensor mount. Keep the breadboard and Arduino in a pouch; avoid loose wiring near moving exercise equipment.

GY-MAX30102: power and I2C check

The front photo identifies the five-pin GY-MAX30102. The back photo is consistent with onboard regulators and level-shifting circuitry, but the component markings/traces do not establish a complete schematic. The page-3 VIN=5V connection is a working assumption for that common board family, not a bench measurement.

With USB unplugged, connect VIN to 5V and GND to GND only; leave SDA/SCL/INT disconnected. Power it and use a multimeter to record VIN and the idle SDA/SCL voltages to GND. For the proposed direct 5V-host connection, SDA and SCL should idle near the host 5V rail. If they idle near 1.8V/3.3V, are unstable, or the layout cannot be verified, stop the direct connection and use a suitable I2C level shifter. A scanner cannot establish voltage compatibility.

After the interface is established, unplug USB; connect SDA to A4 and SCL to A5. Leave INT open. Reconnect and have Person 4 scan for 0x57. If you cannot make the electrical check, finish the MMA7361/timer demo with HR unplugged.

Check	Result / notes
MMA7361 power and awake state	3V3: _____ V Sleep: _____ V GS/ST grounded: _____
Motion response	X/Y/Z change when tilted; no stuck or loose channels: _____
GY power-only readings	VIN: _____ V SDA idle: _____ V SCL idle: _____ V
Interface decision	Direct verified / level shifter required / HR left unplugged: _____
Mounting	Placement and strain relief photo saved: _____

Done when: motion wiring is checked and the HR interface decision is documented.
Owner sign-off: _____ Checked by: _____

Person 4 - program and test A

Name: _____ | Wearable software owner | Work with Person 3

Prepare and upload

Open WearableMonitor/WearableMonitor.ino with PrototypeCore.h beside it. Select Uno R4 Minima; install Arduino UNO R4 Boards. Keep USE_SELECTOR_POT=0. Initially keep ENABLE_MAX30102=0. After Person 3 verifies the HR interface, install SparkFun MAX3010x Pulse and Proximity Sensor Library, set ENABLE_MAX30102=1, and upload again.

Movement and timer demonstration

Use 115200 baud and Newline. While IDLE, send select 5 for lat pulldown, 3 for chest press, 4 for glute bridges or 8 for cat-cow. Fix the sensor placement; send teach and perform three comfortable cycles over 8s. Send go, then compare counts with a human observer. Recalibrate when the placement or profile changes. Use manual while IDLE and rep during RUNNING if the signal is unreliable.

For treadmill, send select 0, then demo: 5s warm-up plus 15s cardio, clearly labelled DEMO. Send real while IDLE to restore the adjustable 180s warm-up plus the required 900s cardio. The button/command go starts, pauses or resumes; finish ends a session.

Test	Expected behavior / evidence
Movement trial	Choose ONE repeatable movement first. Run 3 trials of 10 reps. Record manual/automatic totals; proposed first-demo target: each within +/-1 rep. If missed, report the error and use manual mode.
Pause halfway through rep	No complete count is awarded for a cycle spanning the pause.
Treadmill DEMO	Countdown excluded; warm-up then cardio; DONE after about 20 active seconds. A pause adds no active time.
HR at rest, if connected	After 0x57 is detected, hold a finger steadily on the sensor. Record candidate BPM and compare with a 60-second manual pulse count/reference; record the difference.
No contact / bad signal	BPM is blank with a status, not invented. Finger removal should suppress an old estimate within the next status update.
Baseline and logging	While IDLE, baseline collects 20s. Save wearable.csv with the tested placement, mode and observed failures.

The first HR comparison is an engineering observation, not a clinical accuracy test. Do not create an exercise heart-rate prescription. Select timers/manual confirmation for rotations, airplane, carry and stretches; these sensors do not establish their full technique.

Done when: repeatable movement/timer results are saved; HR is either tested or clearly marked pending.
Owner sign-off: _____ Checked by: _____

References: Arduino R4; MMA7361L; HC-SR04; MAX30102; firsthand GY-board notes; HR library. Schematics are proposed wiring for the photographed parts; physical tests remain the team's work.